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Unit – 4 : Session – 11 : SLO – 1 | Pipeline Hazards — Choose the Best Answer

■ I. Multiple Choice

1. b) Data hazard — dependence on a previous instruction's unfinished result.
2. b) Operand forwarding — routes results directly from one stage to the next, bypassing the register file.
3. c) Control hazard — the pipeline does not know which instruction to fetch until the branch resolves.
4. a) NOP (No Operation) — inserting NOPs (bubbles) stalls the pipeline to handle hazards.
5. c) Structural hazard — two instructions competing for the same functional unit at once.
6. b) Speculative execution (branch prediction) — predicts the branch outcome and proceeds speculatively.
7. d) RAR (Read After Read) — two reads never conflict; RAW, WAR and WAW are the true data hazards.
8. a) Increased CPI — stalls add cycles, raising the average cycles per instruction.
9. a) Fetch — the fetch stage suffers most, since the next PC is unknown until the branch resolves.
10. a) Duplicate functional units — replicated hardware removes the resource conflict causing structural hazards.